

Bahdleen VPN

OpenVPN + WireGuard Installer & Manager

Author: Anthony E. Edjenuwa

Website: edjenuwa.tech

GitHub: github.com/bahdleen

LinkedIn: linkedin.com/in/anthony-e-358226253

This document showcases the installation and usage flow of **Bahdleen VPN**, a personal portfolio project focused on delivering a reliable, user-friendly OpenVPN and WireGuard deployment experience across major Linux distributions.

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1 What This Project Covers

Bahdleen VPN focuses on:

- Cross-distro installation logic for OpenVPN and WireGuard.
- Automated OpenVPN PKI setup (Easy-RSA) and TLS material generation.
- Guided public IP detection with manual fallback.
- DNS selection to push to clients.
- NAT interface detection and persistence.
- Client and peer lifecycle management.
- Export of configs to user-owned directories for clarity and portability.

2 Current Boundaries

This version does not aim to be an enterprise HA platform. It prioritizes a reliable single-server workflow suitable for: portfolio demonstration, lab environments, and production-style proof-of-concepts.

In some environments:

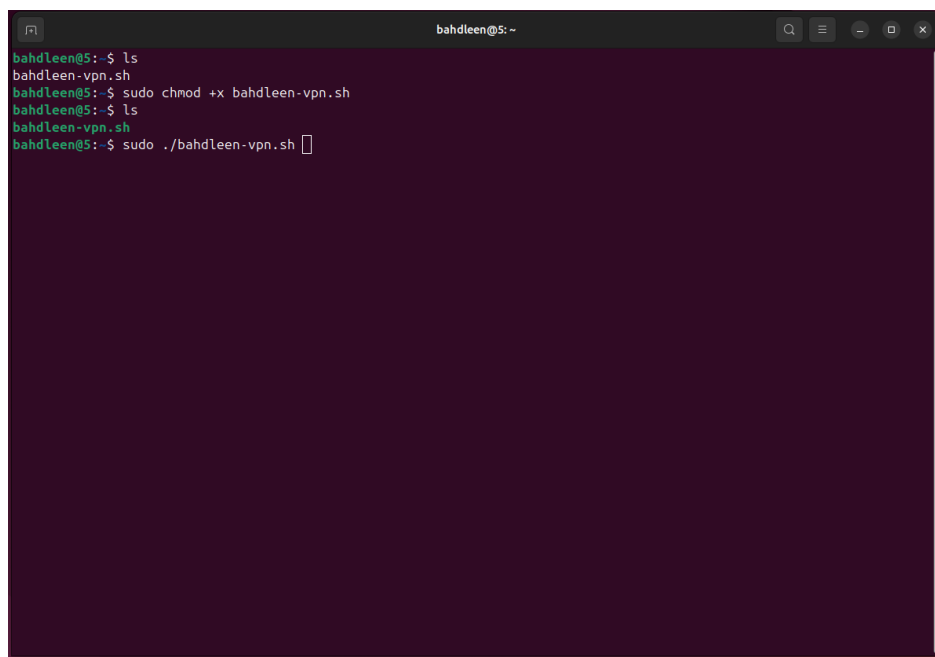
- Public IP lookup can be blocked by provider security.
- Custom firewall baselines may require additional manual rules.
- HA clustering and multi-region failover remain future enhancements.

3 Installation & Usage Walkthrough

This section documents the real installation and setup flow of **Bahdleen VPN** as executed in a live environment. The screenshots are presented in sequence to demonstrate usability, automation logic, and the full deployment lifecycle from first run to client export.

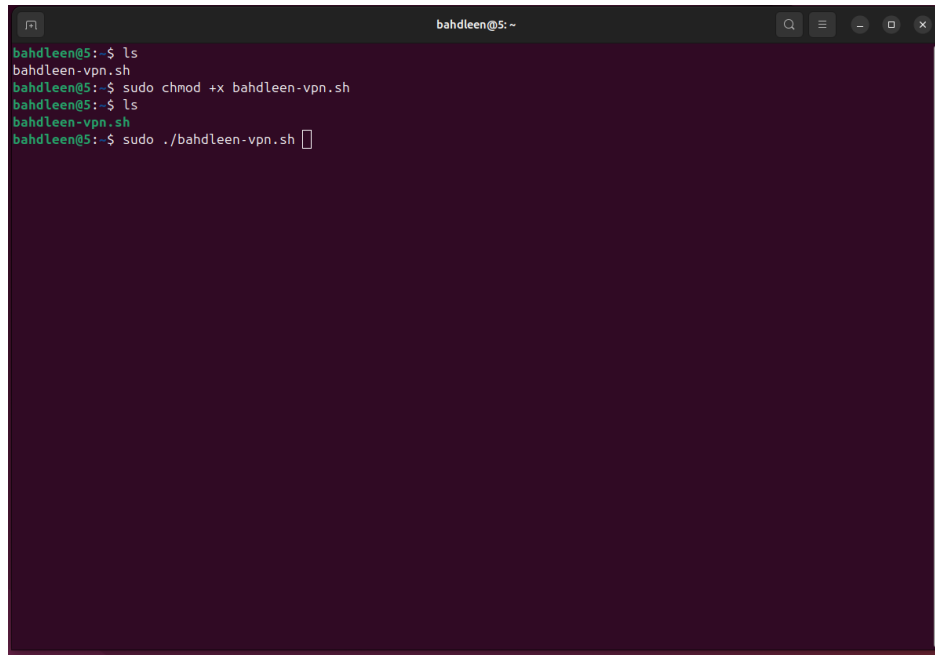
3.1 Getting the Script Ready

The setup begins by confirming the script is present in the working directory, making it executable, and running it with elevated privileges. This ensures the installer can manage packages, services, firewall rules, and VPN interfaces without requiring repeated manual intervention.



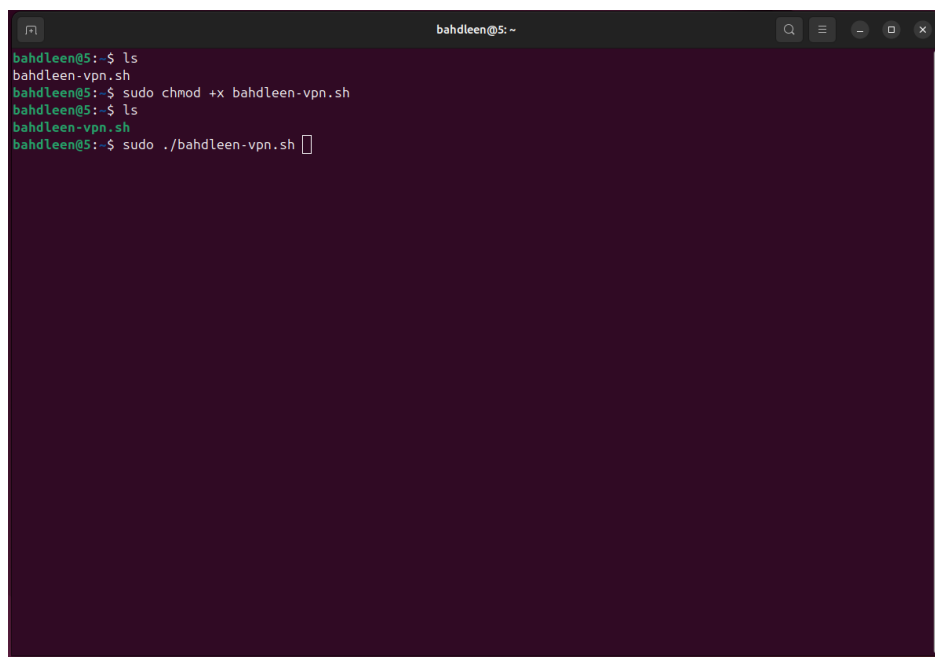
```
bahdleen@5: ~  
bahdleen@5:~$ ls  
bahdleen-vpn.sh  
bahdleen@5:~$ sudo chmod +x bahdleen-vpn.sh  
bahdleen@5:~$ ls  
bahdleen-vpn.sh  
bahdleen@5:~$ sudo ./bahdleen-vpn.sh
```

Figure 1: Listing the project directory to confirm the presence of `bahdleen-vpn.sh`.



```
bahdleen@5: ~  
bahdleen@5:~$ ls  
bahdleen-vpn.sh  
bahdleen@5:~$ sudo chmod +x bahdleen-vpn.sh  
bahdleen@5:~$ ls  
bahdleen-vpn.sh  
bahdleen@5:~$ sudo ./bahdleen-vpn.sh
```

Figure 2: Granting execute permission using `chmod +x`.

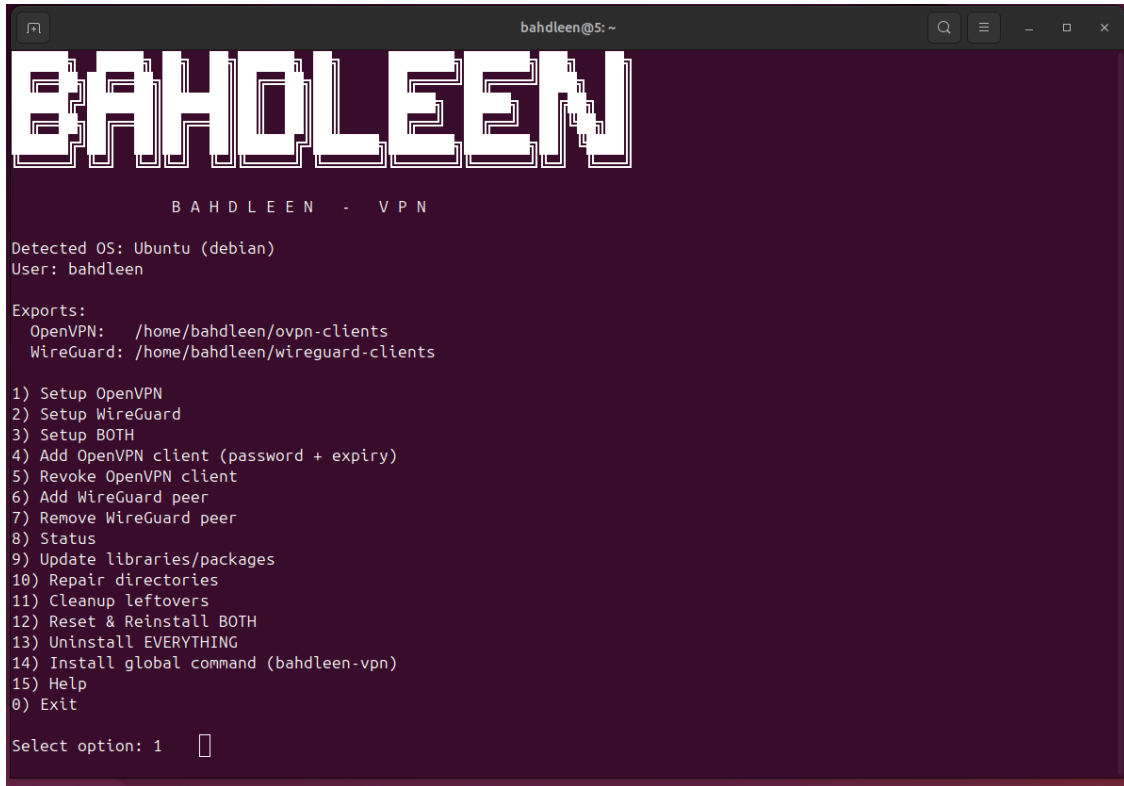


```
bahdleen@5: ~  
bahdleen@5:~$ ls  
bahdleen-vpn.sh  
bahdleen@5:~$ sudo chmod +x bahdleen-vpn.sh  
bahdleen@5:~$ ls  
bahdleen-vpn.sh  
bahdleen@5:~$ sudo ./bahdleen-vpn.sh
```

Figure 3: Launching the script with `sudo ./bahdleen-vpn.sh`.

3.2 Menu Interface and Export Paths

On launch, Bahdleen VPN displays a branded ASCII header alongside system context, including detected OS, active user, and the export destinations for client files. A key design choice is keeping client profiles in user-owned directories rather than hidden system paths.

A terminal window titled 'bahdleen@S: ~' showing the main menu of the Bahdleen VPN application. The menu features a large, stylized ASCII art header 'BAHDLEEN' at the top. Below it, the text 'BAHDLEEN - VPN' is displayed. The terminal then shows system context: 'Detected OS: Ubuntu (debian)' and 'User: bahdleen'. Under the heading 'Exports:', two paths are listed: 'OpenVPN: /home/bahdleen/ovpn-clients' and 'WireGuard: /home/bahdleen/wireguard-clients'. A numbered list of 15 options follows, ranging from '1) Setup OpenVPN' to '0) Exit'. At the bottom, the prompt 'Select option: 1' is shown with a cursor pointing to the number 1.

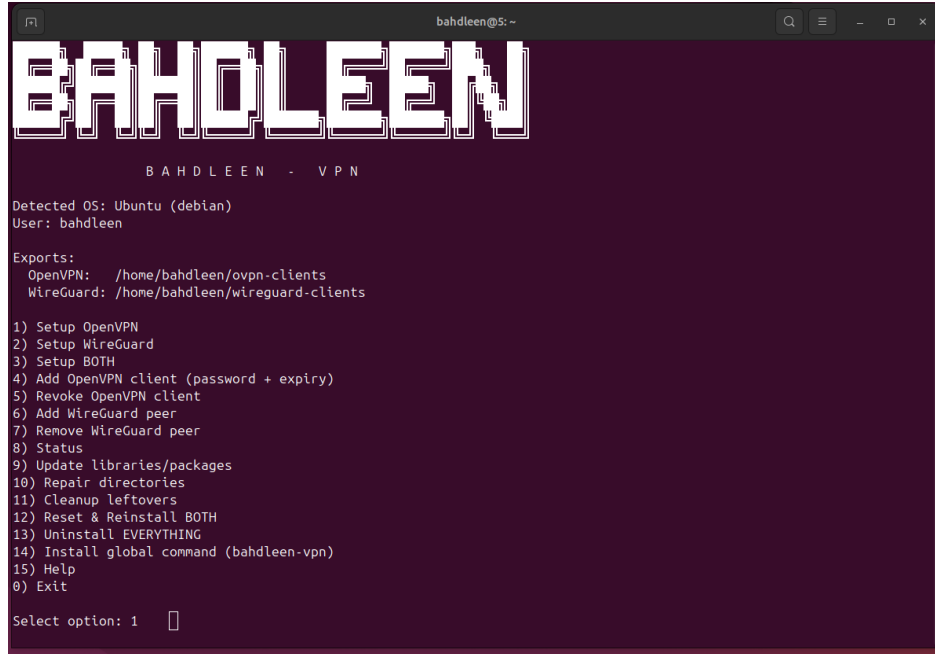
```
bahdleen@S: ~  
BAHDLEEN  
BAHDLEEN - VPN  
Detected OS: Ubuntu (debian)  
User: bahdleen  
Exports:  
OpenVPN: /home/bahdleen/ovpn-clients  
WireGuard: /home/bahdleen/wireguard-clients  
1) Setup OpenVPN  
2) Setup WireGuard  
3) Setup BOTH  
4) Add OpenVPN client (password + expiry)  
5) Revoke OpenVPN client  
6) Add WireGuard peer  
7) Remove WireGuard peer  
8) Status  
9) Update libraries/packages  
10) Repair directories  
11) Cleanup leftovers  
12) Reset & Reinstall BOTH  
13) Uninstall EVERYTHING  
14) Install global command (bahdleen-vpn)  
15) Help  
0) Exit  
Select option: 1
```

Figure 4: The Bahdleen VPN main menu showing detected OS, user, and export locations.

The home-based export model improves portability and reduces confusion during testing, demos, and documentation: `~ovpn-clients` for OpenVPN and `~wireguard-clients` for WireGuard.

3.3 Installing the Global Command

A major usability feature is the ability to install a global launcher so the tool can be reopened later using a single command: `bahdleen-vpn`. This also includes a carefully scoped passwordless sudo rule restricted only to this tool path, which prevents broad escalation while improving user experience.

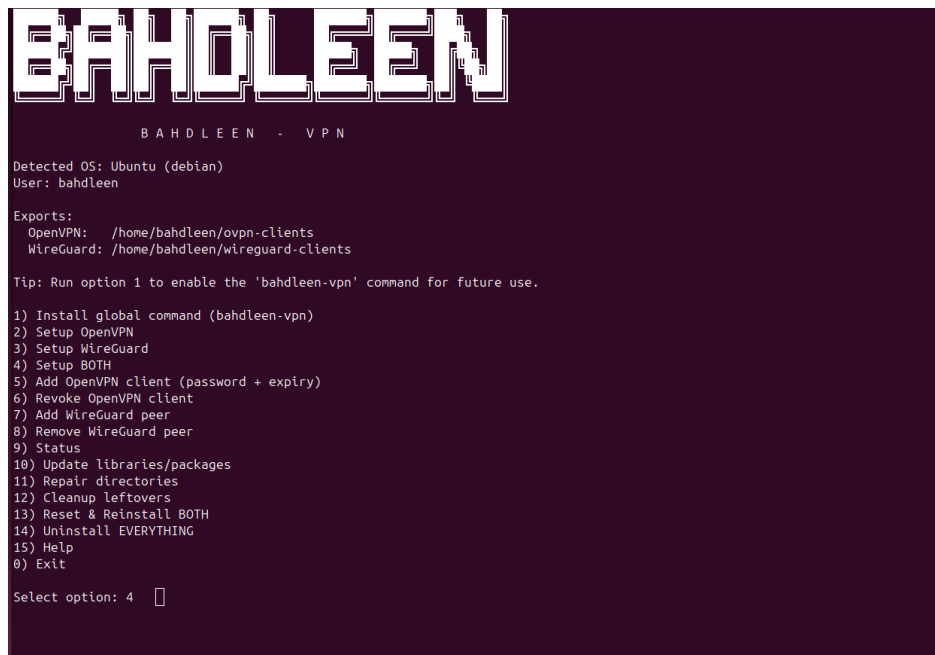


```
bahdleen@S: ~  
BAHDLEEN  
BAHDLEEN - VPN  
Detected OS: Ubuntu (debian)  
User: bahdleen  
Exports:  
  OpenVPN: /home/bahdleen/ovpn-clients  
  WireGuard: /home/bahdleen/wireguard-clients  
1) Setup OpenVPN  
2) Setup WireGuard  
3) Setup BOTH  
4) Add OpenVPN client (password + expiry)  
5) Revoke OpenVPN client  
6) Add WireGuard peer  
7) Remove WireGuard peer  
8) Status  
9) Update libraries/packages  
10) Repair directories  
11) Cleanup leftovers  
12) Reset & Reinstall BOTH  
13) Uninstall EVERYTHING  
14) Install global command (bahdleen-vpn)  
15) Help  
0) Exit  
Select option: 1
```

Figure 5: Global command install prompt showing the script path, launcher path, and restricted sudo policy.

3.4 Running a Combined Deployment (OpenVPN + WireGuard)

The tool supports independent or combined installation. In this walkthrough, the combined option is used to validate the full automation workflow and NAT consistency across both VPN services.

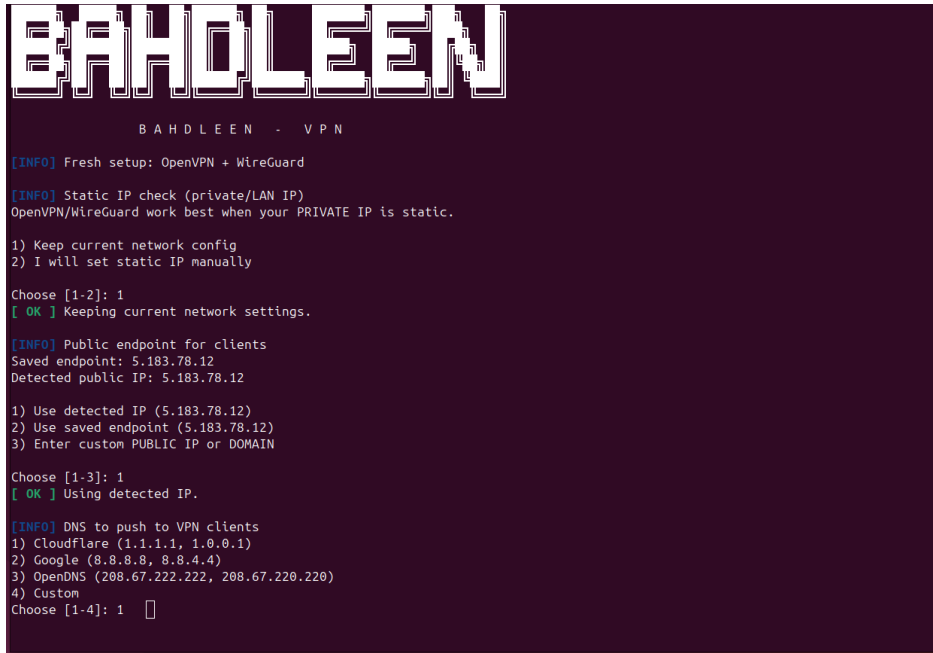


```
BAHDLEEN  
BAHDLEEN - VPN  
Detected OS: Ubuntu (debian)  
User: bahdleen  
Exports:  
  OpenVPN: /home/bahdleen/ovpn-clients  
  WireGuard: /home/bahdleen/wireguard-clients  
Tip: Run option 1 to enable the 'bahdleen-vpn' command for future use.  
1) Install global command (bahdleen-vpn)  
2) Setup OpenVPN  
3) Setup WireGuard  
4) Setup BOTH  
5) Add OpenVPN client (password + expiry)  
6) Revoke OpenVPN client  
7) Add WireGuard peer  
8) Remove WireGuard peer  
9) Status  
10) Update libraries/packages  
11) Repair directories  
12) Cleanup leftovers  
13) Reset & Reinstall BOTH  
14) Uninstall EVERYTHING  
15) Help  
0) Exit  
Select option: 4
```

Figure 6: Selecting the combined setup option for OpenVPN + WireGuard.

3.5 Static Private IP Advisory

Before deployment proceeds, the script includes a short private/LAN IP advisory. This does not force configuration changes but promotes best practice by warning that VPN stability is improved when the server's private IP is static.



```
BAHDLEEN
BAHDLEEN - VPN

[INFO] Fresh setup: OpenVPN + WireGuard

[INFO] Static IP check (private/LAN IP)
OpenVPN/WireGuard work best when your PRIVATE IP is static.

1) Keep current network config
2) I will set static IP manually

Choose [1-2]: 1
[ OK ] Keeping current network settings.

[INFO] Public endpoint for clients
Saved endpoint: 5.183.78.12
Detected public IP: 5.183.78.12

1) Use detected IP (5.183.78.12)
2) Use saved endpoint (5.183.78.12)
3) Enter custom PUBLIC IP or DOMAIN

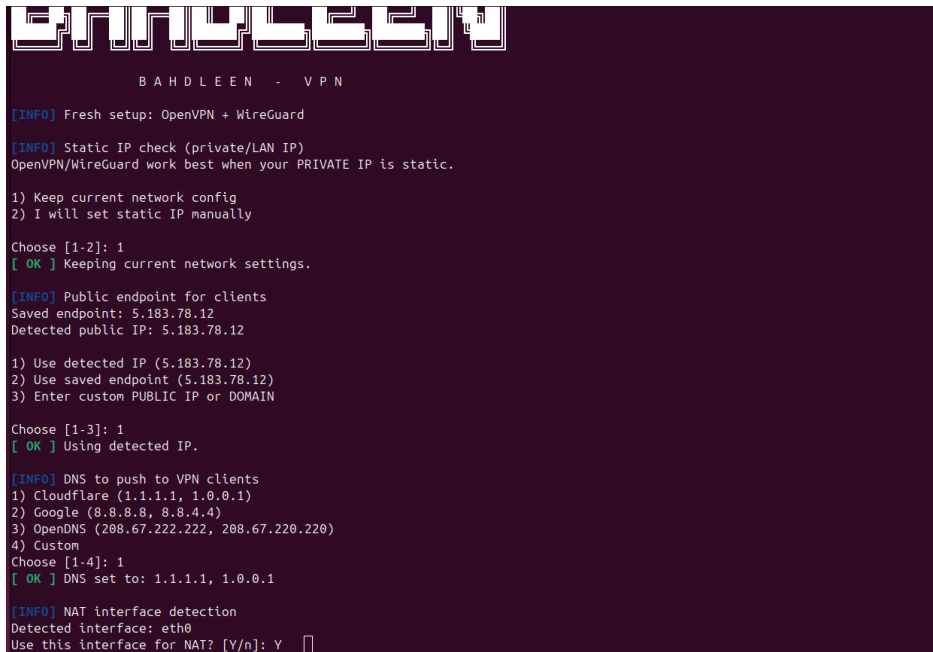
Choose [1-3]: 1
[ OK ] Using detected IP.

[INFO] DNS to push to VPN clients
1) Cloudflare (1.1.1.1, 1.0.0.1)
2) Google (8.8.8.8, 8.8.4.4)
3) OpenDNS (208.67.222.222, 208.67.220.220)
4) Custom
Choose [1-4]: 1
```

Figure 7: Private/LAN IP advisory with options to keep current settings or set static manually.

3.6 Public Endpoint Detection with Safe Fallback

A common issue in many VPN installers is unreliable or blocked public IP detection. Bahdleen VPN implements a resilient approach: it detects the public IP when possible, shows it to the user, and also offers a saved endpoint or manual input option.



```
BAHDLEEN
BAHDLEEN - VPN

[INFO] Fresh setup: OpenVPN + WireGuard

[INFO] Static IP check (private/LAN IP)
OpenVPN/WireGuard work best when your PRIVATE IP is static.

1) Keep current network config
2) I will set static IP manually

Choose [1-2]: 1
[ OK ] Keeping current network settings.

[INFO] Public endpoint for clients
Saved endpoint: 5.183.78.12
Detected public IP: 5.183.78.12

1) Use detected IP (5.183.78.12)
2) Use saved endpoint (5.183.78.12)
3) Enter custom PUBLIC IP or DOMAIN

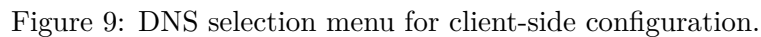
Choose [1-3]: 1
[ OK ] Using detected IP.

[INFO] DNS to push to VPN clients
1) Cloudflare (1.1.1.1, 1.0.0.1)
2) Google (8.8.8.8, 8.8.4.4)
3) OpenDNS (208.67.222.222, 208.67.220.220)
4) Custom
Choose [1-4]: 1
[ OK ] DNS set to: 1.1.1.1, 1.0.0.1

[INFO] NAT interface detection
Detected interface: eth0
Use this interface for NAT? [Y/n]: Y
```

Figure 8: Public IP detection flow showing saved endpoint and detected endpoint options.

To keep the deployment suitable for real usage, the script prompts for DNS preferences that will be pushed to VPN clients. This provides a clear balance between convenience and control.



Because NAT is required for common remote-access VPN scenarios, the tool detects the primary egress interface and asks for confirmation before applying persistent rules.



3.9 OpenVPN PKI and Cryptographic Setup

The OpenVPN setup process includes fully automated PKI initialization using Easy-RSA. This covers server certificates, TLS material, and Diffie-Hellman parameters.

[illegible]

Figure 11: OpenVPN PKI generation and certificate/database initialization, the key generation takes longer depending on your machine resource.

[illegible]

Figure 12: DH parameter generation and TLS crypt key production.

3.10 Successful Dual-VPN Completion

Once OpenVPN and WireGuard are installed, Bahdleen VPN applies NAT rules that cover both services and confirms that each system component has been configured successfully.

```
.....+.....
.....+.....++*++*++*++*

DH parameters of size 2048 created at /etc/openvpn/easy-rsa/pki/dh.pem

Note: using Easy-RSA configuration from: ./vars

Using SSL: openssl OpenSSL 1.1.1f  31 Mar 2020
Using configuration from /etc/openvpn/easy-rsa/pki/safessl-easyrsa.cnf

An updated CRL has been created.
CRL file: /etc/openvpn/easy-rsa/pki/crl.pem

[ OK ] PKI initialized.
[INFO] Generating tls-crypt key...
[ OK ] TLS key generated.
[INFO] Installing OpenVPN server keys/certs...
[ OK ] Server crypto installed.

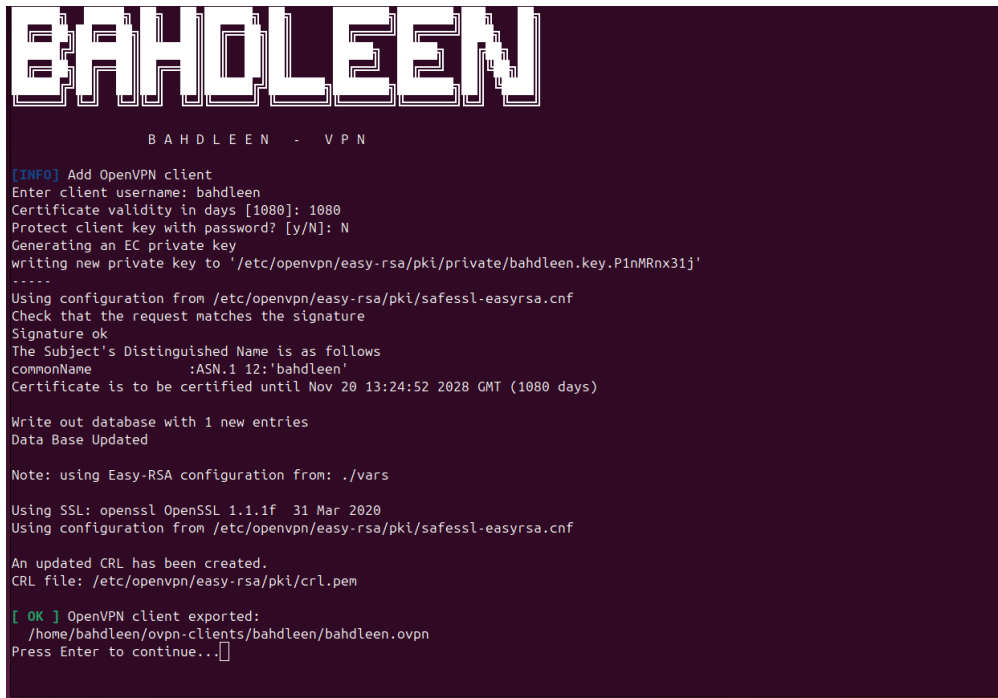
OpenVPN port [1194]: 1194
Protocol udp/tcp [udp]: udp
[INFO] Writing OpenVPN server config...
[ OK ] Server config created.
[ OK ] OpenVPN service started (bahdleen-openvpn).
[ OK ] OpenVPN setup complete.
[INFO] Setting up WireGuard...
[INFO] Generating WireGuard server keys...
[ OK ] WireGuard server keys ready.

WireGuard port [51820]: 51820
[INFO] Writing WireGuard server config...
[ OK ] WireGuard server config created.
[ OK ] WireGuard service started (wg-quick@wg0).
[ OK ] WireGuard setup complete.
[INFO] Applying NAT rules for OpenVPN + WireGuard...
[ OK ] NAT configured and persistent.
[ OK ] Both VPNs installed and configured.
Press Enter to continue...[ ]
```

Figure 13: Combined setup completion showing OpenVPN + WireGuard configured and NAT applied.

3.11 Adding an OpenVPN Client Profile

Client provisioning is intentionally guided and flexible. The tool supports username input, certificate validity control, and optional passphrase protection for private keys.



```
BAHDLEEN
BAHDLEEN - VPN

[INFO] Add OpenVPN client
Enter client username: bahdleen
Certificate validity in days [1080]: 1080
Protect client key with password? [y/N]: N
Generating an EC private key
writing new private key to '/etc/openvpn/easy-rsa/pki/private/bahdleen.key.P1nMRnx31j'
-----
Using configuration from /etc/openvpn/easy-rsa/pki/safessl-easyrsa.cnf
Check that the request matches the signature
Signature ok
The Subject's Distinguished Name is as follows
commonName      :ASN.1 12:'bahdleen'
Certificate is to be certified until Nov 20 13:24:52 2028 GMT (1080 days)

Write out database with 1 new entries
Data Base Updated

Note: using Easy-RSA configuration from: ./vars

Using SSL: openssl OpenSSL 1.1.1f  31 Mar 2020
Using configuration from /etc/openvpn/easy-rsa/pki/safessl-easyrsa.cnf

An updated CRL has been created.
CRL file: /etc/openvpn/easy-rsa/pki/crl.pem

[ OK ] OpenVPN client exported:
/home/bahdleen/ovpn-clients/bahdleen/bahdleen.ovpn
Press Enter to continue...

```

Figure 14: OpenVPN client creation prompt: username, certificate validity, and optional password protection.

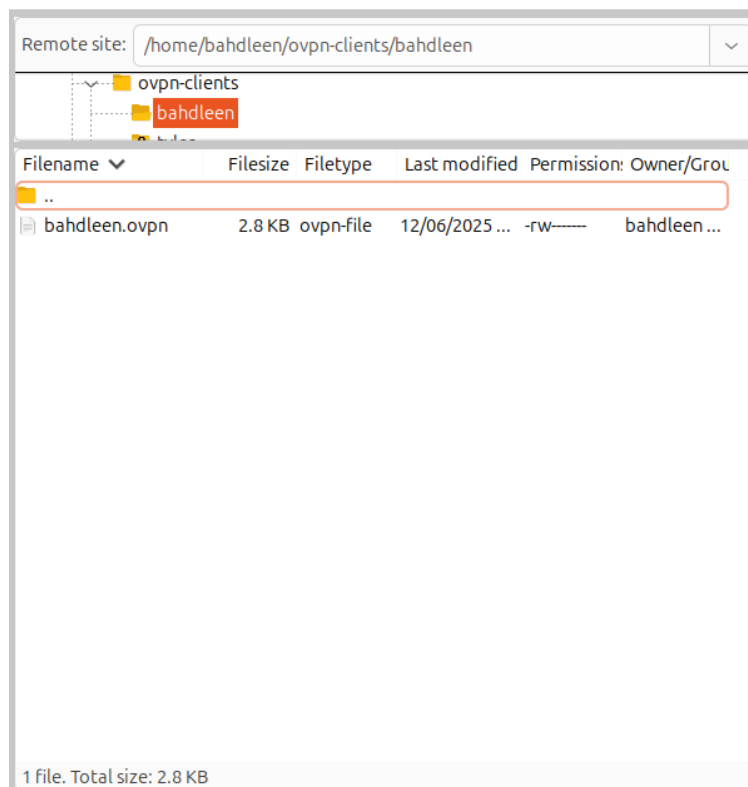


Figure 15: Successful export of the OpenVPN client configuration into `/home/bahdleen/ovpn-clients`.

3.12 Reopening the Tool After Installation

After the global command is installed, Bahdleen VPN can be launched from anywhere on the system without returning to the project directory. This improves daily usability and makes the tool feel like a native utility rather than a one-time script.

From this point onward, `bahdleen-vpn` is the key command to start the tool.

```
bahdleen-vpn
```

With this workflow, administrators can repeatedly add or revoke clients without restarting the entire setup process, supporting practical operational usage.

This sequence forms a complete and reproducible installation narrative suitable for documentation, portfolio evidence, and a step-by-step video demonstration.